

TW1.1 - Git Workflow & Collaboration Report

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class:CSE B

Subject: DevOps Lab

Aim

To understand distributed version control using Git and GitHub and perform repository creation, cloning, branching, committing, merging and pushing changes.

Objectives

- Learn Git basics.
- Work with local and remote repositories.
- Understand branching and merging.
- Collaborate using GitHub.

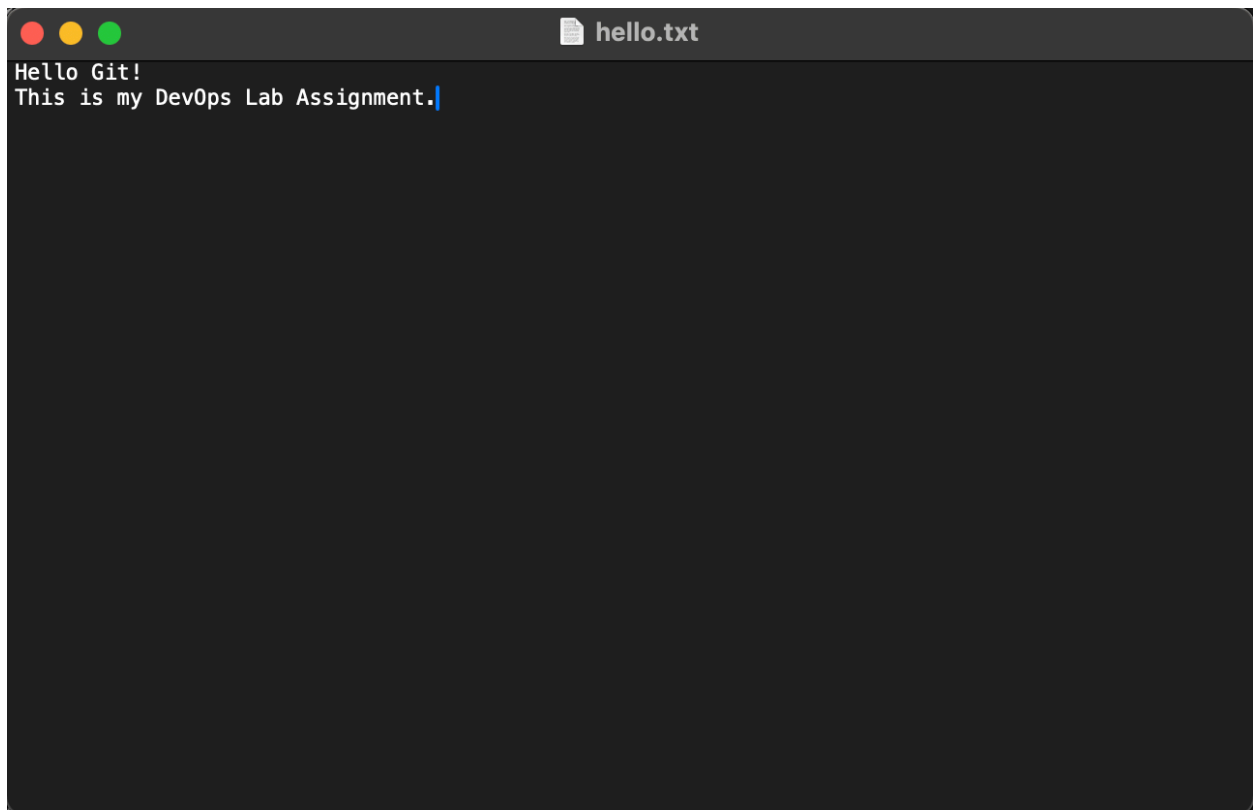
Theory

Git is a distributed version control system developed by Linus Torvalds. It allows developers to track changes in source code, maintain version history, collaborate efficiently, resolve merge conflicts, and manage software projects. GitHub is a cloud platform that hosts Git repositories and enables collaboration through pull requests, issues, and repository management.

Software Requirements

Git, GitHub account, Windows/Linux, Internet connection, VS Code.

```
jyotisah@JYOTIs-MacBook-Air ~ %  
[Restored 30 Jul 2026 at 12:33:36 AM]  
Last login: Fri Jul 24 14:43:10 on console  
jyotisah@JYOTIs-MacBook-Air ~ % clear  
  
jyotisah@JYOTIs-MacBook-Air ~ % git --version  
git version 2.39.5 (Apple Git-154)  
jyotisah@JYOTIs-MacBook-Air ~ % git config --global user.name  
jyotisah@JYOTIs-MacBook-Air ~ % git config --global user.email  
jyotisah@JYOTIs-MacBook-Air ~ % git config --global user.name "Jyoti Kumari Sah"  
jyotisah@JYOTIs-MacBook-Air ~ % git config --global user.email "jyotisah413@gmail.com"  
jyotisah@JYOTIs-MacBook-Air ~ % mkdir GitWorkflowLab  
jyotisah@JYOTIs-MacBook-Air ~ % cd GitWorkflowLab  
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % pwd  
/Users/jyotisah/GitWorkflowLab  
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git init  
Initialized empty Git repository in /Users/jyotisah/GitWorkflowLab/.git/  
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % touch hello.txt  
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % open hello.txt  
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % Hello Git!  
[This is my DevOps Lab Assignment.  
zsh: command not found: Hello  
zsh: command not found: This  
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % open hello.txt  
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab %
```



```

jyotisah@JYOTIs-MacBook-Air ~ %
[Restored 30 Jul 2026 at 12:33:36 AM]
Last login: Fri Jul 24 14:43:10 on console
jyotisah@JYOTIs-MacBook-Air ~ % clear

jyotisah@JYOTIs-MacBook-Air ~ % git --version
git version 2.39.5 (Apple Git-154)
jyotisah@JYOTIs-MacBook-Air ~ % git config --global user.name
jyotisah@JYOTIs-MacBook-Air ~ % git config --global user.email
jyotisah@JYOTIs-MacBook-Air ~ % git config --global user.name "Jyoti Kumari Sah"
jyotisah@JYOTIs-MacBook-Air ~ % git config --global user.email "jyotisah413@gmail.com"
jyotisah@JYOTIs-MacBook-Air ~ % mkdir GitWorkflowLab
cd GitWorkflowLab
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % pwd
/Users/jyotisah/GitWorkflowLab
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git init
Initialized empty Git repository in /Users/jyotisah/GitWorkflowLab/.git/
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % touch hello.txt
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % open hello.txt
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % Hello Git!
This is my DevOps Lab Assignment.
zsh: command not found: Hello
zsh: command not found: This
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % open hello.txt
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.txt

nothing added to commit but untracked files present (use "git add" to track)
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git add .
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   hello.txt

jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git commit -m "Initial commit"
[main (root-commit) 105d25d] Initial commit
 1 file changed, 2 insertions(+)
 create mode 100644 hello.txt

```

git branch, git checkout, git merge

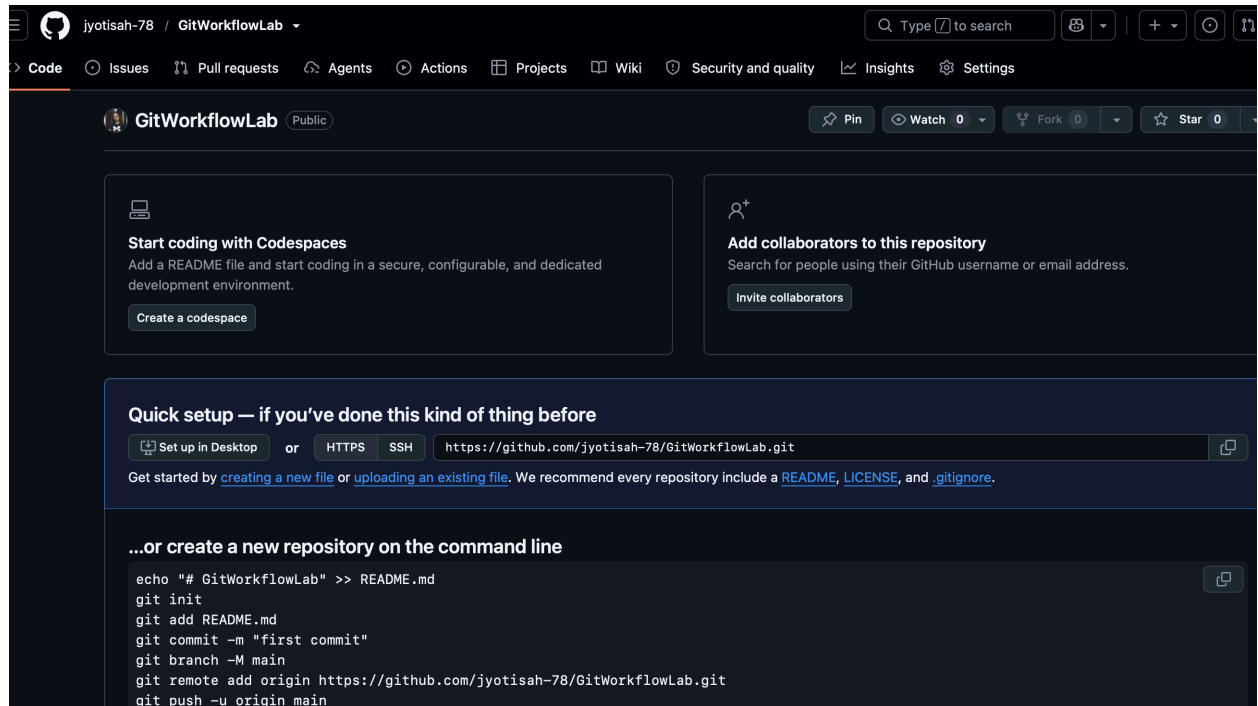
```

    create mode 100644 hello.txt
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git branch feature
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git branch
  feature
* main
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git checkout feature
Switched to branch 'feature'
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % open hello.txt
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git status
On branch feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   hello.txt

no changes added to commit (use "git add" and/or "git commit -a")
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git add .
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git commit -m "Updated hello.txt in feature branch"
[feature 4ccd0ad] Updated hello.txt in feature branch
 1 file changed, 2 insertions(+), 1 deletion(-)
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git checkout main
Switched to branch 'main'
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git merge feature
Updating 105d25d..4ccd0ad
Fast-forward
 hello.txt | 3 ++-
 1 file changed, 2 insertions(+), 1 deletion(-)
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git log --oneline
4ccd0ad (HEAD -> main, feature) Updated hello.txt in feature branch
105d25d Initial commit

```

, git push, git pull.



```
no changes added to commit (use "git add" and/or "git commit -a")
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git add .
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git commit -m "Updated hello.txt in feature branch"
[feature 4ccd0ad] Updated hello.txt in feature branch
1 file changed, 2 insertions(+), 1 deletion(-)
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git checkout main
Switched to branch 'main'
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git merge feature
Updating 105d25d..4ccd0ad
Fast-forward
 hello.txt | 3 ++-
1 file changed, 2 insertions(+), 1 deletion(-)
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git log --oneline
4ccd0ad (HEAD -> main, feature) Updated hello.txt in feature branch
105d25d Initial commit
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git remote add origin https://github.com/jyotisah-78/GitWorkflowLab.git
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git branch -M main
jyotisah@JYOTIs-MacBook-Air GitWorkflowLab % git push -u origin main
```


Commands Used

git init, git clone, git status, git add ., git commit -m, git branch, git checkout, git merge, git push, git pull.

Learning Outcomes

Learned version control, repository management, collaborative development, and basic Git workflow.

Conclusion

Git simplifies version control and enables efficient collaboration while maintaining complete project history.